

Telco Customer Churn : Executive Overview

1 in 4 customers churns, putting nearly half of monthly revenue at risk. Monthly contracts are the main risk factor .

Churn Rate

26,54 %

High Risk Customers

Model prediction

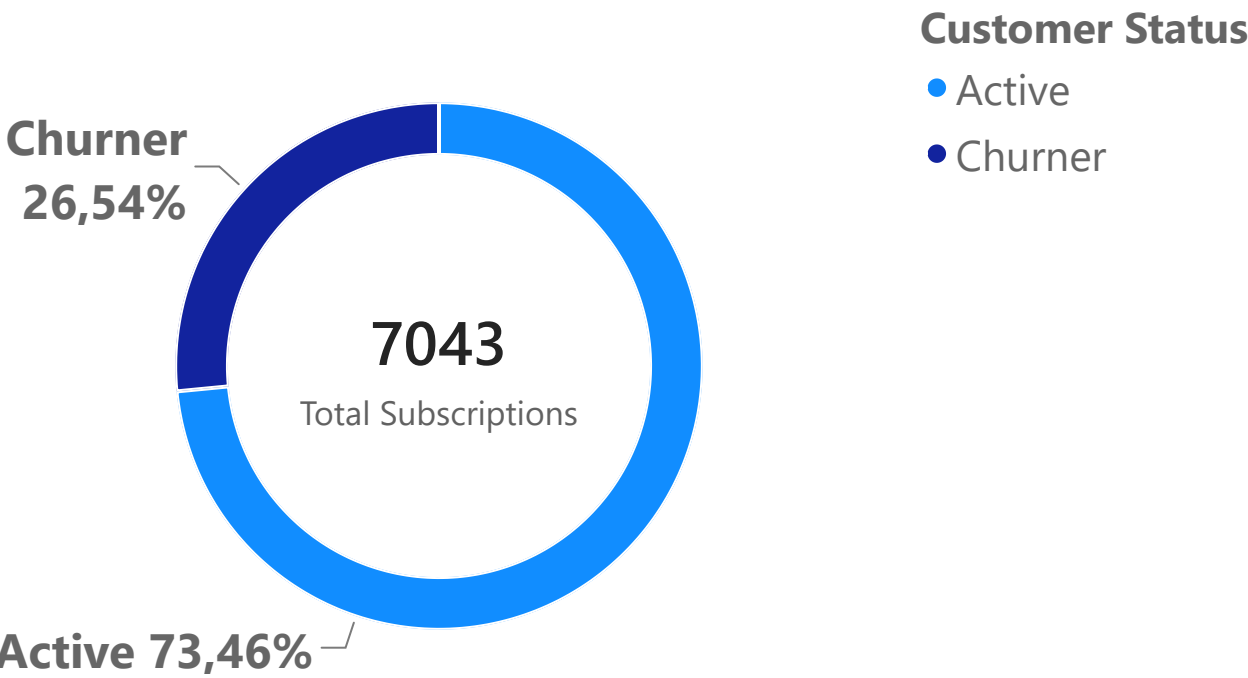
1646

Revenue at Risk

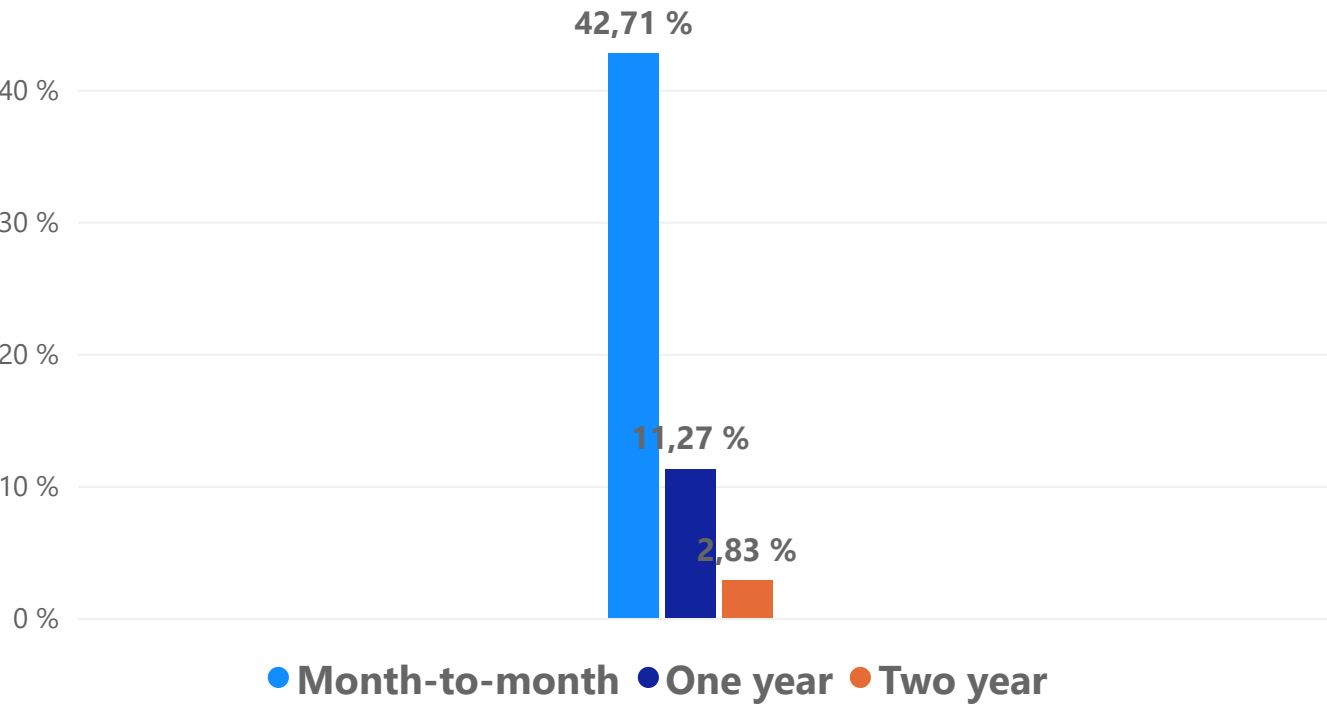
If predicted churners leave

\$223 738

Total Subscriptions by Customer Status



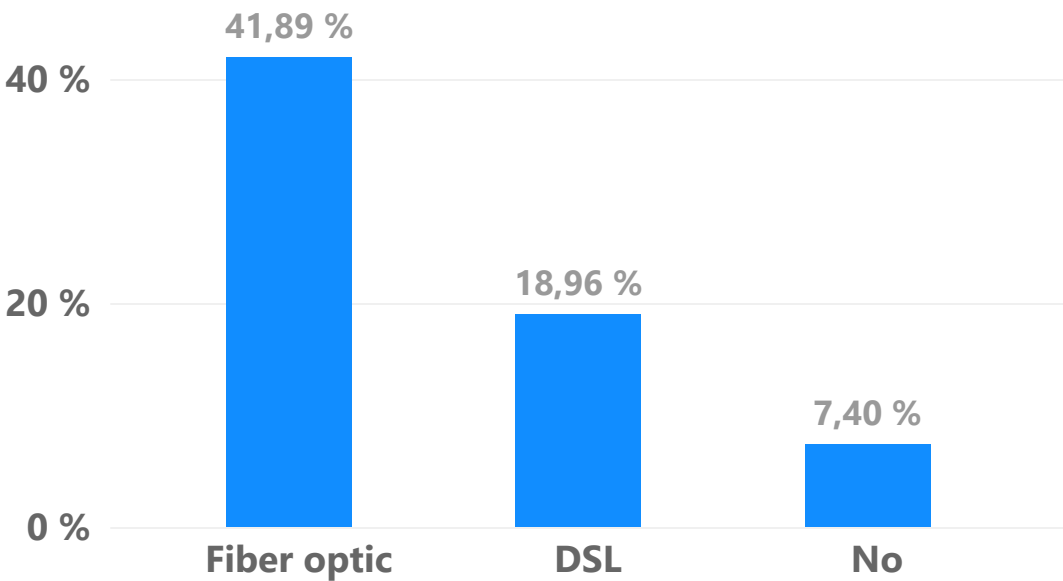
Churn Rate by type of Contract



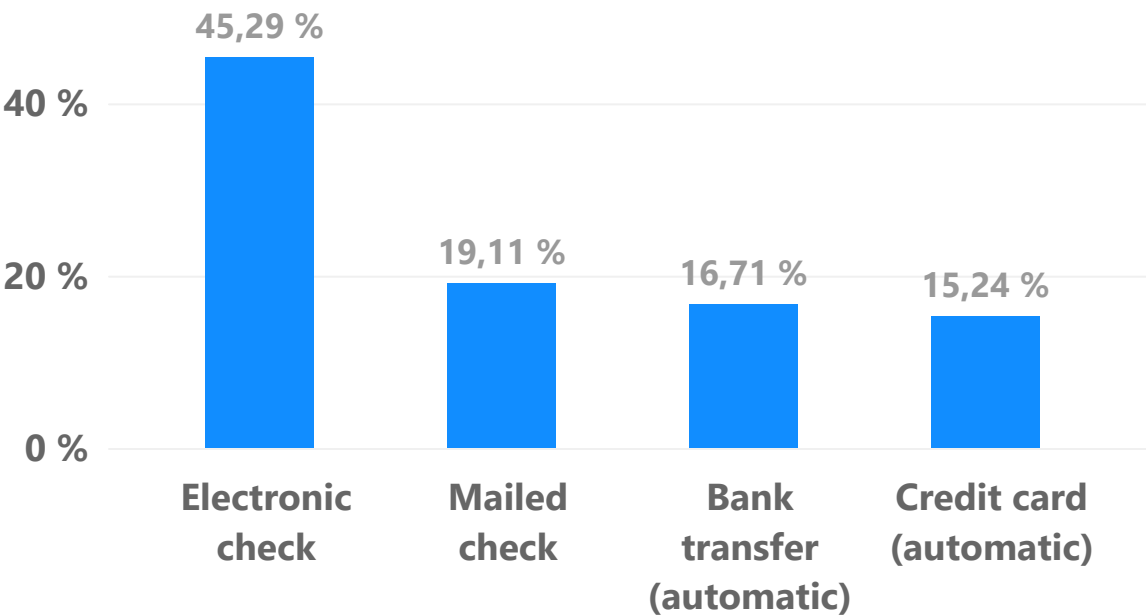
Churns Drivers

Several churn drivers emerged from the analysis. Our goal is to identify the most impactful ones to act on, and to check if a causal relationship exists.

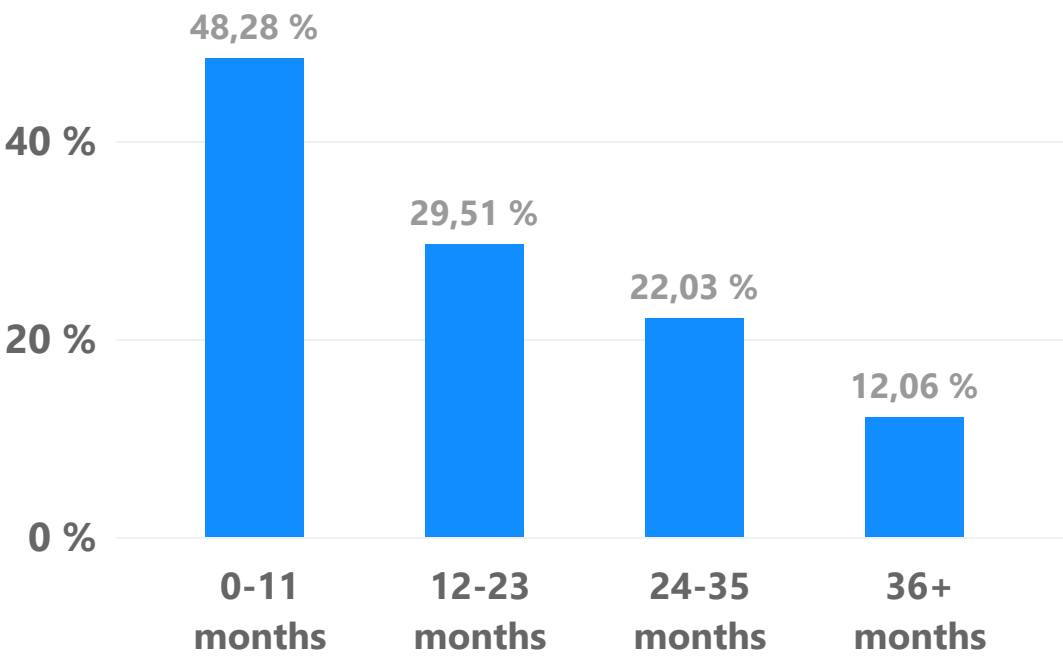
Churn Rate by Internet Service



Churn Rate by Payment Method



Churn Rate by Tenure

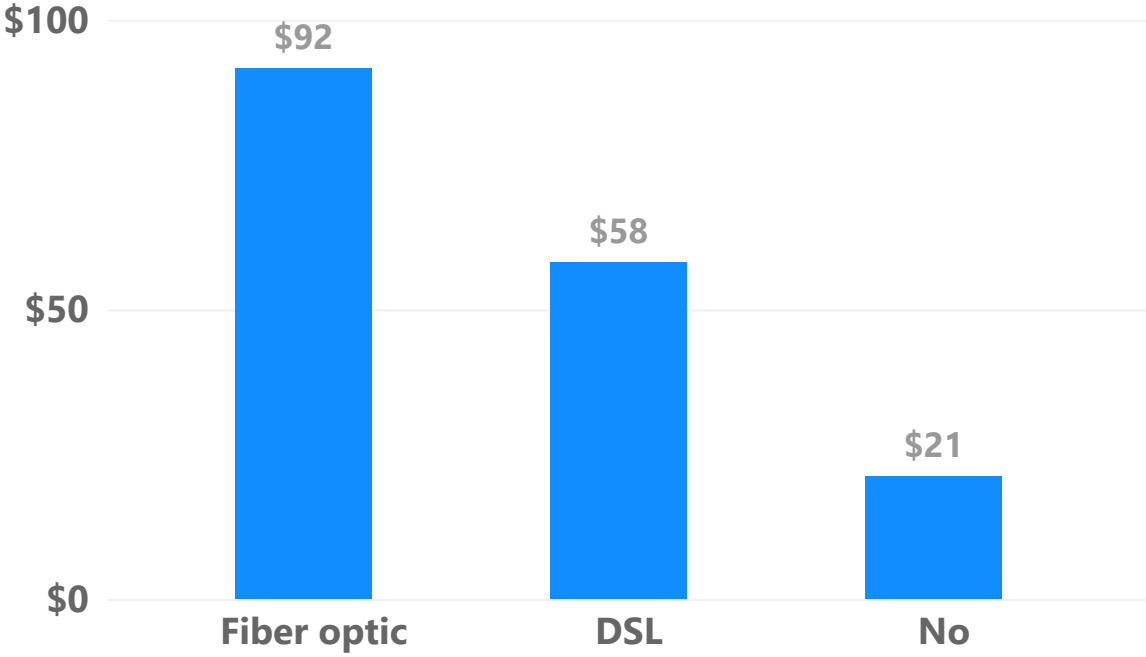


Churn Rate by Contract × Internet Service

Month-to-month + fiber optic is the highest risk combination

Contract	No	DSL	Fiber optic
Month-to-month	18,89 %	32,22 %	54,61 %
One year	2,47 %	9,30 %	19,29 %
Two year	0,78 %	1,91 %	7,23 %

Average of Monthly Charges by Internet Service



Model Insights

These results come from the prediction model (XGBOOST).The metrics below are measured on the test set , customers the model never saw during training.

Recall
Detects 8 out of 10 real churners

81,0 %

Precision
About half of the alerts are true churners

50,6 %

F1
A balanced score between detection and precision

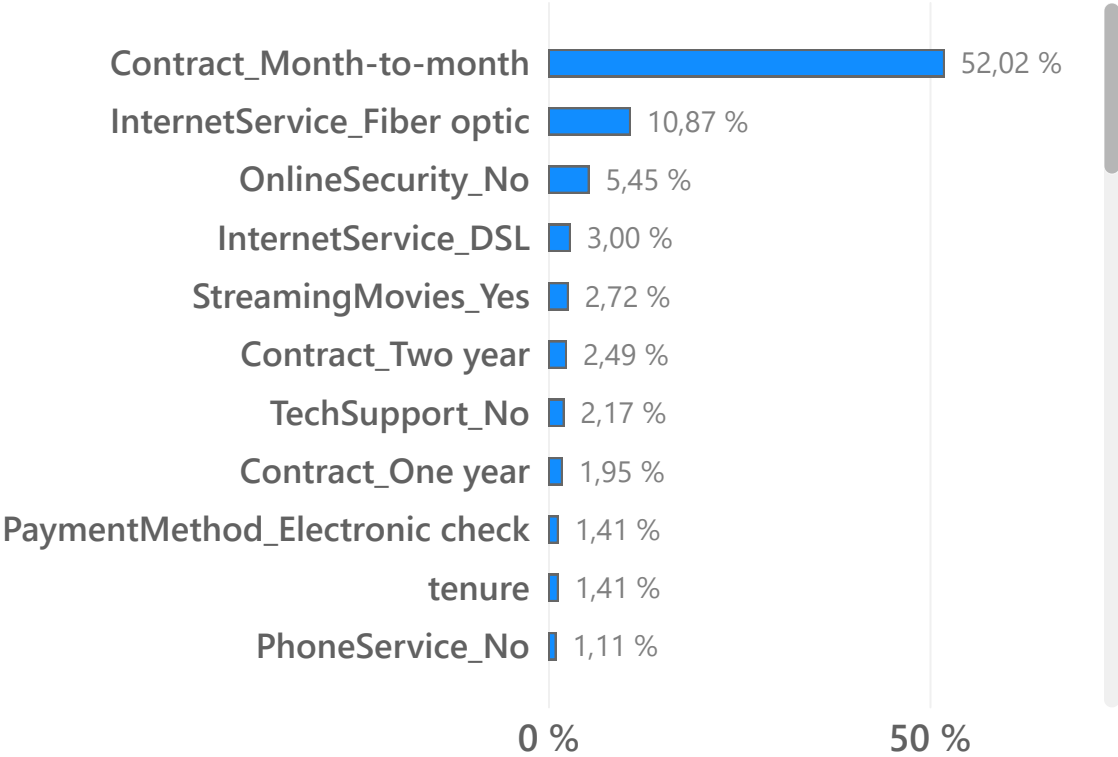
62,3 %

ROC-AUC
Correctly ranks churners vs non-churners in 84% of cases

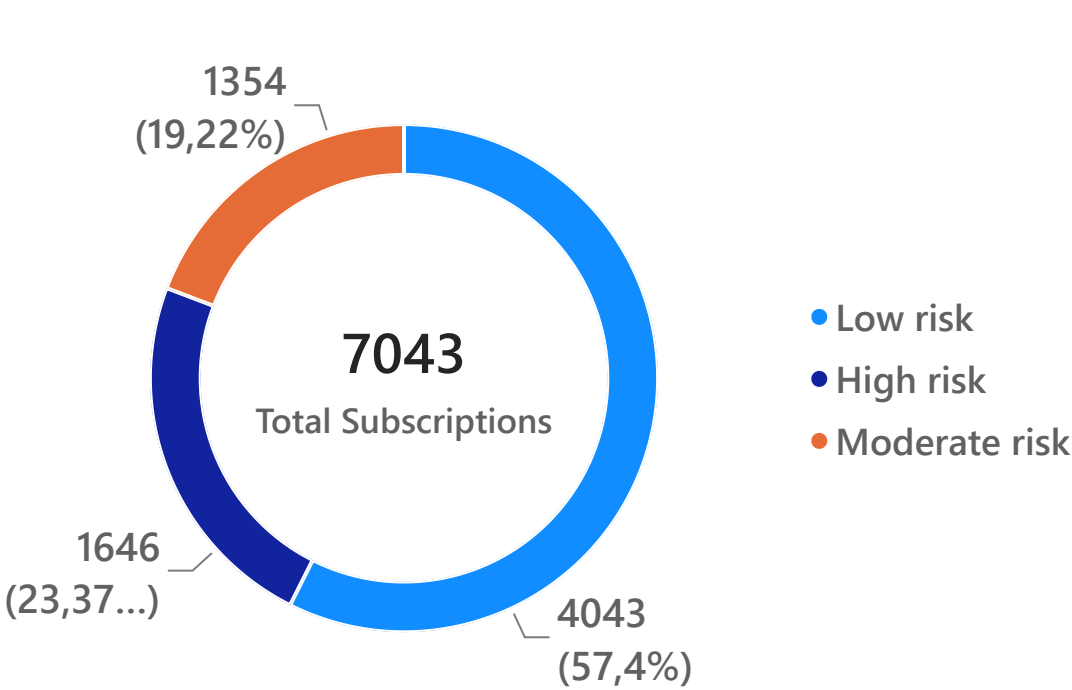
83,9 %

Top Features Driving Churn

Contract type dominates, 5x more than the second feature

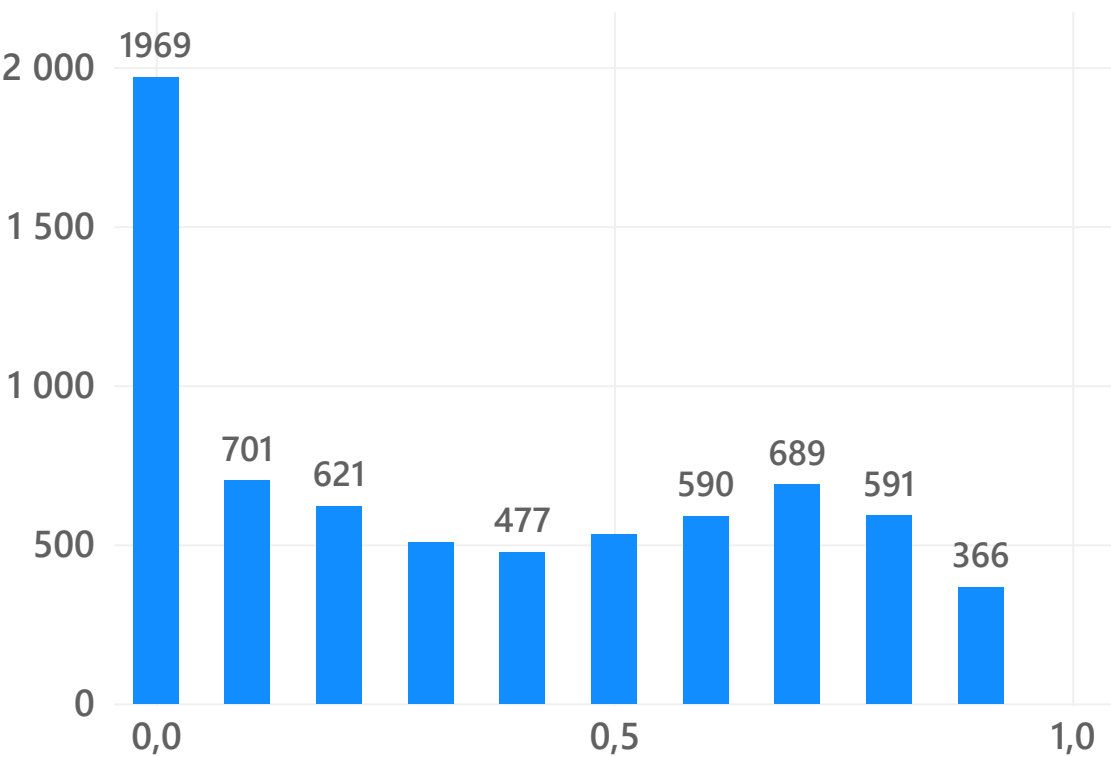


Total Subscriptions by Risk Segment



Distribution of Churn Probabilities

Most customers have low probability, a core group is at high risk



Business Recommendations

The Model identifies who to act on. Here is what to do, and on whom first.

Contract

All

Risk Segment

All

InternetService

All

Customers to Contact First

Sorted by churn probability, highest first

customerID	Churn_Probability	Contract	Sum of MonthlyCharges	Sum of tenure	Risk Segment
5178-LMXOP	97,13 %	Month-to-month	95,10	1	High risk
9300-AGZNL	97,08 %	Month-to-month	94,00	1	High risk
9497-QCMMS	97,08 %	Month-to-month	93,55	1	High risk
7216-EWTRS	96,86 %	Month-to-month	100,80	1	High risk
4910-GMJOT	96,76 %	Month-to-month	94,60	1	High risk
8149-RSOUN	96,76 %	Month-to-month	93,85	1	High risk
0295-PPHDO	96,61 %	Month-to-month	95,45	1	High risk
0107-YHINA	96,55 %	Month-to-month	99,75	1	High risk
5419-JPRRN	96,48 %	Month-to-month	101,45	1	High risk
1069-XAIEM	96,38 %	Month-to-month	85,05	1	High risk
3068-OMWZA	96,15 %	Month-to-month	88,80	1	High risk
4826-XTSOH	96,15 %	Month-to-month	86,05	1	High risk
5192-EBGOV	96,15 %	Month-to-month	85,70	1	High risk
6894-LFHLY	96,09 %	Month-to-month	75,75	1	High risk

- Month-to-month customers churn at 42.7% vs 2.8% for two-year contracts. Offer incentives to switch to annual Contracts.
- Fiber optic churns at 41.9% and costs 57% more than DSL. Address the price/value perception.
- Online security reduces churn risk. Bundle it with retention offers.
- 75% of churners leave before month 29. Act early in the customer journey, not after.

Recoverable Revenue

Assuming a 20% retention rate on at-risk revenue

\$44 747,60

Acting on monthly contracts and Fiber optic pricing could protect a significant share of the 223.738 \$ at risk.